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manchodu@Srini:~$ cd OSLAB
manchodu@Srini:~/OSLAB$ nano history.c
manchodu@Srini:~/OSLAB$ gcc history.c -o history
manchodu@Srini:~/OSLAB$ ./history
Simple Command History Program
Type commands and press Enter.
Use UP/DOWN arrows to navigate history.
Type 'exit' to quit.

OSLAB> |
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Simple Command History Program
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OSLAB> hi
Command entered: hi
OSLAB> bye
Command entered: bye
OSLAB> |
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manchodu@Sriini:~/OSLAB$ cat history.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define MAX_HISTORY 10
#define MAX_BUFFER 100

char history[MAX_HISTORY][MAX_BUFFER];
int history_count = 0;
int history_index = 0;

struct termios original_terminal;

/* Restore normal terminal settings */
void disable_raw_mode()
{
    tcsetattr(STDIN_FILENO, TCSAFLUSH, &original_terminal);
}

/* Enable character-by-character input */
void enable_raw_mode()
{
    tcgetattr(STDIN_FILENO, &original_terminal);
    atexit(disable_raw_mode);

    struct termios raw = original_terminal;

    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

/* Add command to history */
void add_history(char *command)
{
    if (strlen(command) == 0)
        return;

    if (history_count < MAX_HISTORY)
    {

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    if (history_count < MAX_HISTORY)
    {
        strcpy(history[history_count], command);
        history_count++;
    }
    else
    {
        for (int i = 0; i < MAX_HISTORY - 1; i++)
        {
            strcpy(history[i], history[i + 1]);
        }

        strcpy(history[MAX_HISTORY - 1], command);
    }

    history_index = history_count;
}

/* Clear current input line */
void clear_line(int length)
{
    for (int i = 0; i < length; i++)
        printf("\b \b");

    fflush(stdout);
}

/* Read input with history support */
void read_command(char *buffer)
{
    int position = 0;
    int length = 0;

    history_index = history_count;

    while (1)
    {
        char c;

        read(STDIN_FILENO, &c, 1);

        /* Enter */
        if (c == '\n' || c == '\r')
        {
            buffer[length] = '\0';
            printf("\n");
            return;
        }
    }
}

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/* Backspace */
if (c == 127 || c == 8)
{
    if (length > 0)
    {
        length--;
        position--;

        printf("\b \b");
        fflush(stdout);
    }

    continue;
}

/* Escape sequence */
if (c == 27)
{
    char seq[2];

    read(STDIN_FILENO, &seq[0], 1);
    read(STDIN_FILENO, &seq[1], 1);

    /* UP arrow */
    if (seq[0] == '[' && seq[1] == 'A')
    {
        if (history_index > 0)
        {
            history_index--;

            clear_line(length);

            strcpy(buffer, history[history_index]);

            length = strlen(buffer);
            position = length;

            printf("%s", buffer);
            fflush(stdout);
        }
    }

    /* DOWN arrow */
    else if (seq[0] == '[' && seq[1] == 'B')
    {
        if (history_index < history_count - 1)

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/* DOWN arrow */
else if (seq[0] == '[' && seq[1] == 'B')
{
    if (history_index < history_count - 1)
    {
        history_index++;

        clear_line(length);

        strcpy(buffer, history[history_index]);

        length = strlen(buffer);
        position = length;

        printf("%s", buffer);
        fflush(stdout);
    }
    else if (history_index == history_count - 1)
    {
        history_index++;

        clear_line(length);

        buffer[0] = '\0';
        length = 0;
        position = 0;
    }
}

continue;
}

/* Normal character */
if (c >= 32 && c <= 126)
{
    if (length < MAX_BUFFER - 1)
    {
        buffer[position] = c;
        position++;
        length++;

        buffer[length] = '\0';

        printf("%c", c);
        fflush(stdout);
    }
}

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        /* Normal character */
        if (c >= 32 && c <= 126)
        {
            if (length < MAX_BUFFER - 1)
            {
                buffer[position] = c;
                position++;
                length++;

                buffer[length] = '\0';

                printf("%c", c);
                fflush(stdout);
            }
        }
    }
}

int main()
{
    char buffer[MAX_BUFFER];

    enable_raw_mode();

    printf("Simple Command History Program\n");
    printf("Type commands and press Enter.\n");
    printf("Use UP/DOWN arrows to navigate history.\n");
    printf("Type 'exit' to quit.\n\n");

    while (1)
    {
        printf("OSLAB> ");
        fflush(stdout);

        read_command(buffer);

        if (strcmp(buffer, "exit") == 0)
            break;

        add_history(buffer);

        printf("Command entered: %s\n", buffer);
    }

    return 0;
}

```